

Assessing the Network with Common Security Tools (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 01

Student:

Christopher Phan

Email:

cphan@udallas.edu

Time on Task:

31 hours, 3 minutes

Progress:

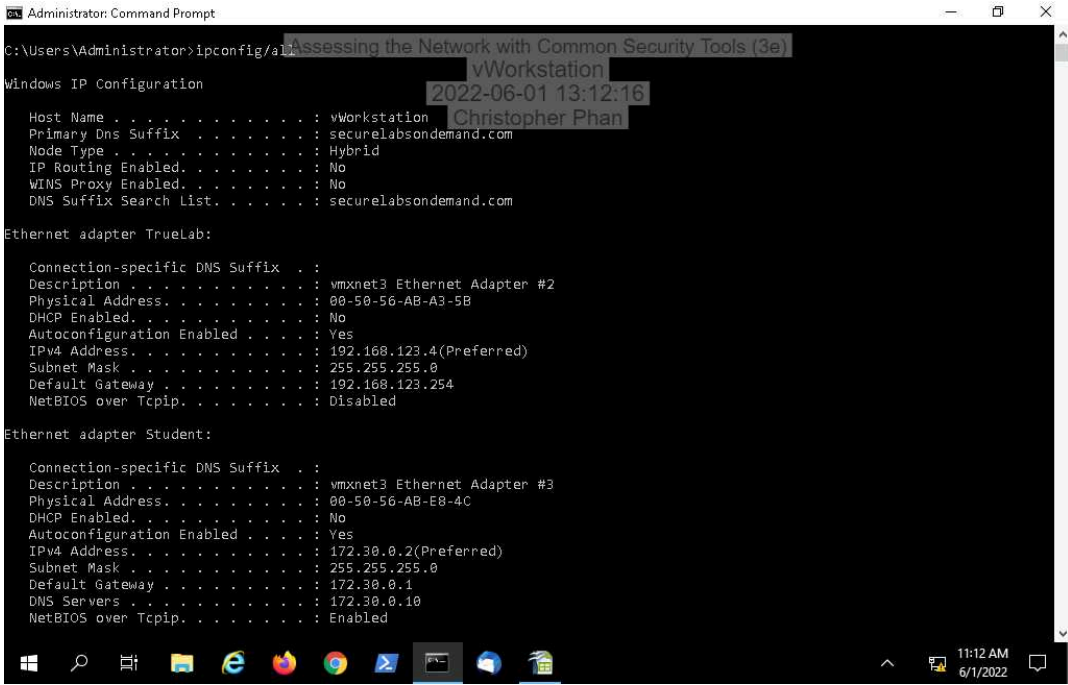
100%

Report Generated: Tuesday, June 7, 2022 at 12:28 PM

Section 1: Hands-On Demonstration

Part 1: Explore the Local Area Network

4. Make a screen capture showing the **ipconfig** results for the Student adapter on the vWorkstation.



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The command entered is `ipconfig /all`. The output displays network configuration for the vWorkstation. The "Ethernet adapter TrueLab:" section shows an IPv4 address of 192.168.123.4. The "Ethernet adapter Student:" section shows an IPv4 address of 172.30.0.2. The window has a taskbar at the bottom with various application icons and a system clock showing 11:12 AM on 6/1/2022.

```
C:\Users\Administrator>ipconfig /all

Windows IP Configuration

Host Name . . . . . : vWorkstation
Primary Dns Suffix . . . . . : securelabsondemand.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : securelabsondemand.com

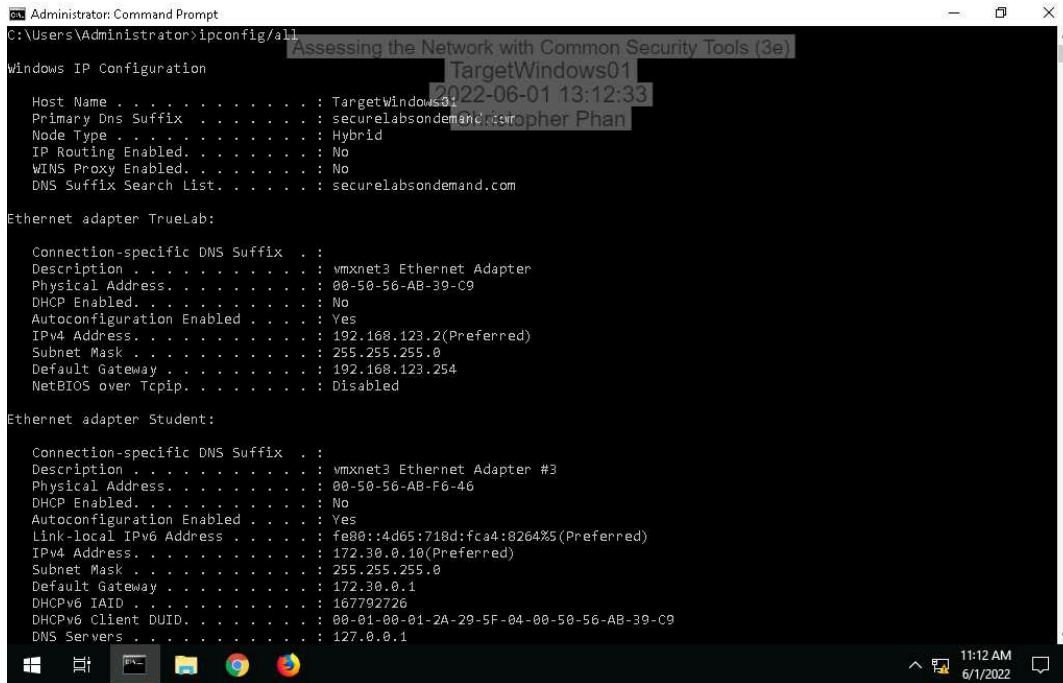
Ethernet adapter TrueLab:

   Connection-specific DNS Suffix  . : 
   Description . . . . . : vmxnet3 Ethernet Adapter #2
   Physical Address. . . . . : 00-50-56-AB-A3-5B
   DHCP Enabled. . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . : 192.168.123.4(Preferred)
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.123.254
   NetBIOS over Tcpip. . . . . : Disabled

Ethernet adapter Student:

   Connection-specific DNS Suffix  . : 
   Description . . . . . : vmxnet3 Ethernet Adapter #3
   Physical Address. . . . . : 00-50-56-AB-E8-4C
   DHCP Enabled. . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . : 172.30.0.2(Preferred)
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 172.30.0.1
   DNS Servers . . . . . : 172.30.0.10
   NetBIOS over Tcpip. . . . . : Enabled
```

7. Make a screen capture showing the **ipconfig** results for the Student adapter on **TargetWindows01**.



```
Administrator Command Prompt
C:\Users\Administrator>ipconfig/all

Windows IP Configuration

Host Name . . . . . : TargetWindows01
Primary Dns Suffix . . . . . : securelabsondemand.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : securelabsondemand.com

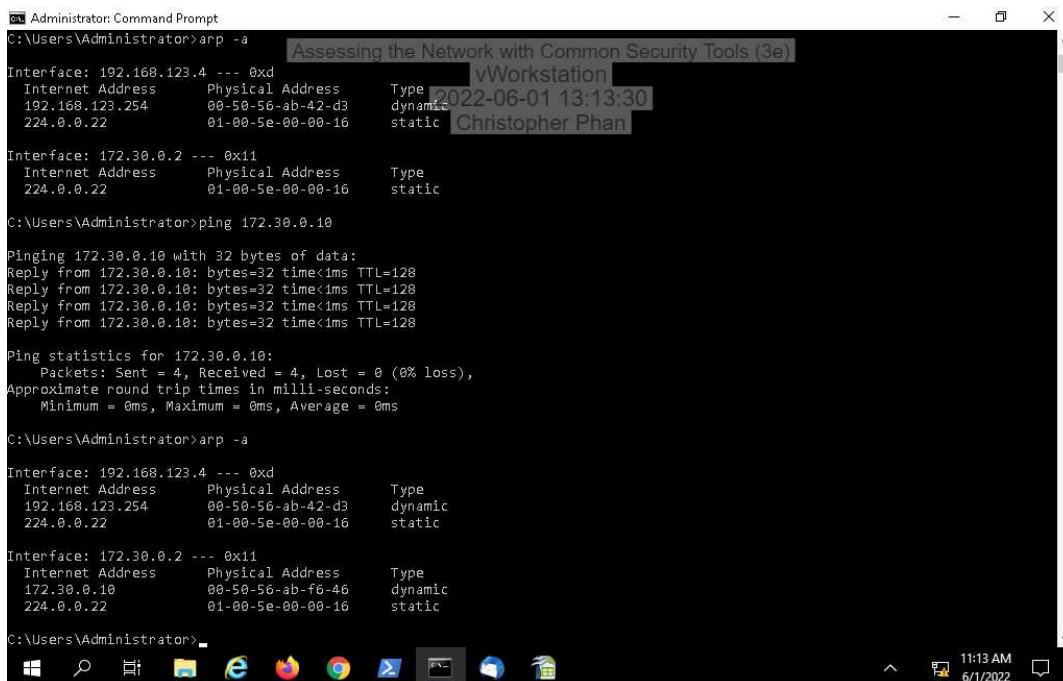
Ethernet adapter TrueLab:

Connection-specific DNS Suffix . : 
Description . . . . . : vmxnet3 Ethernet Adapter
Physical Address. . . . . : 00-50-56-AB-39-C9
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.123.2(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.123.254
NetBIOS over Tcpip. . . . . : Disabled

Ethernet adapter Student:

Connection-specific DNS Suffix . : 
Description . . . . . : vmxnet3 Ethernet Adapter #3
Physical Address. . . . . : 00-50-56-AB-F6-46
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . : fe80::4d65:718d:fca4:8264%5(Preferred)
IPv4 Address. . . . . : 172.30.0.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 172.30.0.1
DHCPv6 IAD . . . . . : 167792726
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-29-5F-04-00-50-56-AB-39-C9
DNS Servers . . . . . : 127.0.0.1
```

15. Make a screen capture showing the updated ARP cache on the **vWorkstation**.



```
Administrator Command Prompt
C:\Users\Administrator>arp -a

Interface: 192.168.123.4 --- 0xd
Internet Address      Physical Address      Type
192.168.123.254       00-50-56-ab-42-d3     dynamic
224.0.0.22            01-00-5e-00-00-16     static

Interface: 172.30.0.2 --- 0x11
Internet Address      Physical Address      Type
224.0.0.22            01-00-5e-00-00-16     static

C:\Users\Administrator>ping 172.30.0.10

Pinging 172.30.0.10 with 32 bytes of data:
Reply from 172.30.0.10: bytes=32 time<1ms TTL=128
Reply from 172.30.0.10: bytes=32 time<1ms TTL=128
Reply from 172.30.0.10: bytes=32 time<1ms TTL=128
Reply from 172.30.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 172.30.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>arp -a

Interface: 192.168.123.4 --- 0xd
Internet Address      Physical Address      Type
192.168.123.254       00-50-56-ab-42-d3     dynamic
224.0.0.22            01-00-5e-00-00-16     static

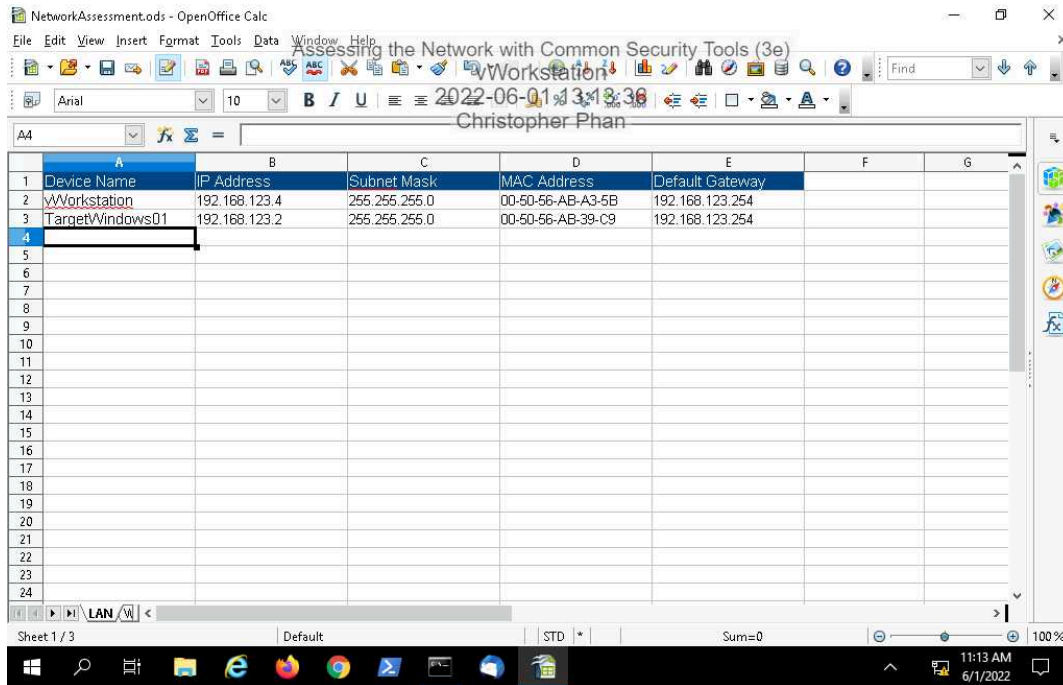
Interface: 172.30.0.2 --- 0x11
Internet Address      Physical Address      Type
172.30.0.10           00-50-56-ab-f6-46     dynamic
224.0.0.22            01-00-5e-00-00-16     static

C:\Users\Administrator>
```

Assessing the Network with Common Security Tools (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 01

19. Make a screen capture showing the **completed LAN tab** of the Network Assessment spreadsheet.

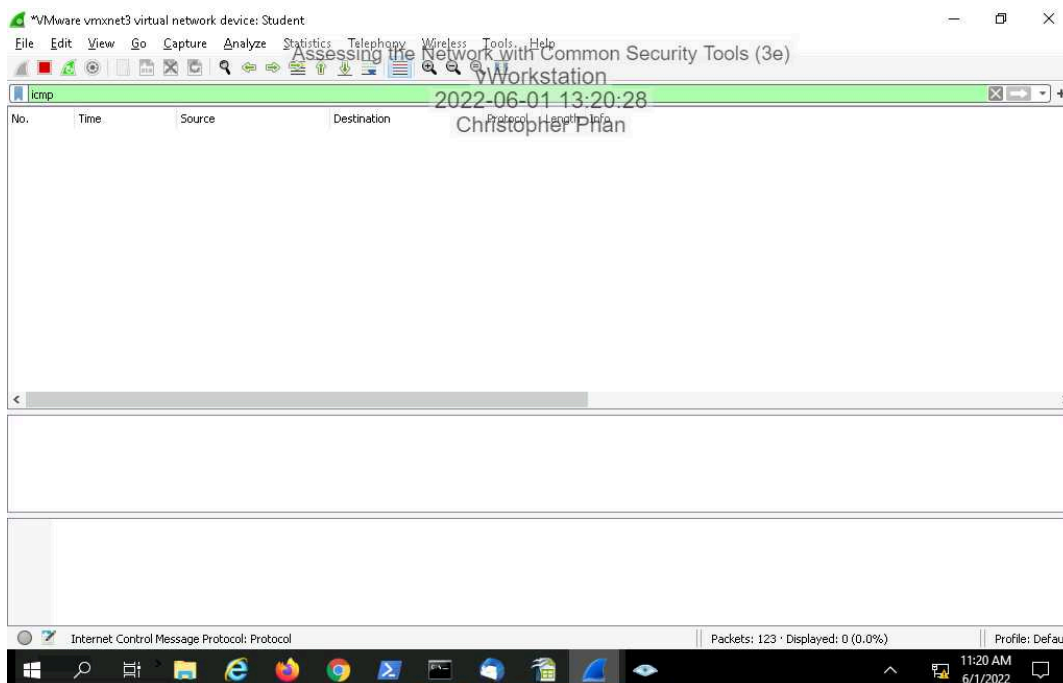


The screenshot shows the OpenOffice Calc application window titled "NetworkAssessment.ods - OpenOffice Calc". The spreadsheet is titled "Assessing the Network with Common Security Tools (3e)" and is the work of "Christopher Phan". The "LAN" tab is selected. The spreadsheet contains the following data:

Device Name	IP Address	Subnet Mask	MAC Address	Default Gateway
Workstation	192.168.123.4	255.255.255.0	00-50-56-AB-A3-5B	192.168.123.254
TargetWindows01	192.168.123.2	255.255.255.0	00-50-56-AB-39-C9	192.168.123.254

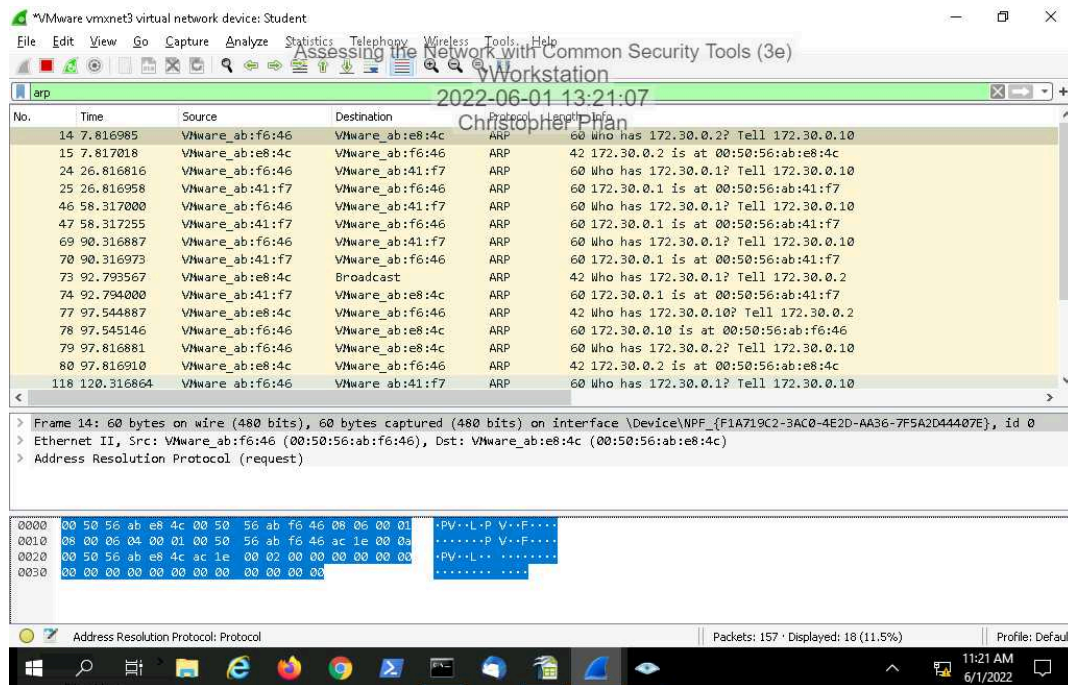
Part 2: Analyze Network Traffic

9. Make a screen capture showing the **ICMP filtered results** in Wireshark.



The screenshot shows the Wireshark application window titled "*VMware vmxnet3 virtual network device: Student". The "Filter" bar is set to "icmp". The packet list shows 123 packets, with 0 displayed. The packet details pane shows the "Internet Control Message Protocol: Protocol" section. The status bar at the bottom indicates "Packets: 123 · Displayed: 0 (0.0%)".

12. Make a screen capture showing the ARP filtered results in Wireshark.



18. Compare the Regular scan results for ICMP and ARP traffic with the results from the Ping scan.

The Ping scan simply shows the MAC address and how many hosts on within the network through their IP address. While the Regular scan shows all this as well as showing all the TCP ports connected to the network, and their types such as services and their status.

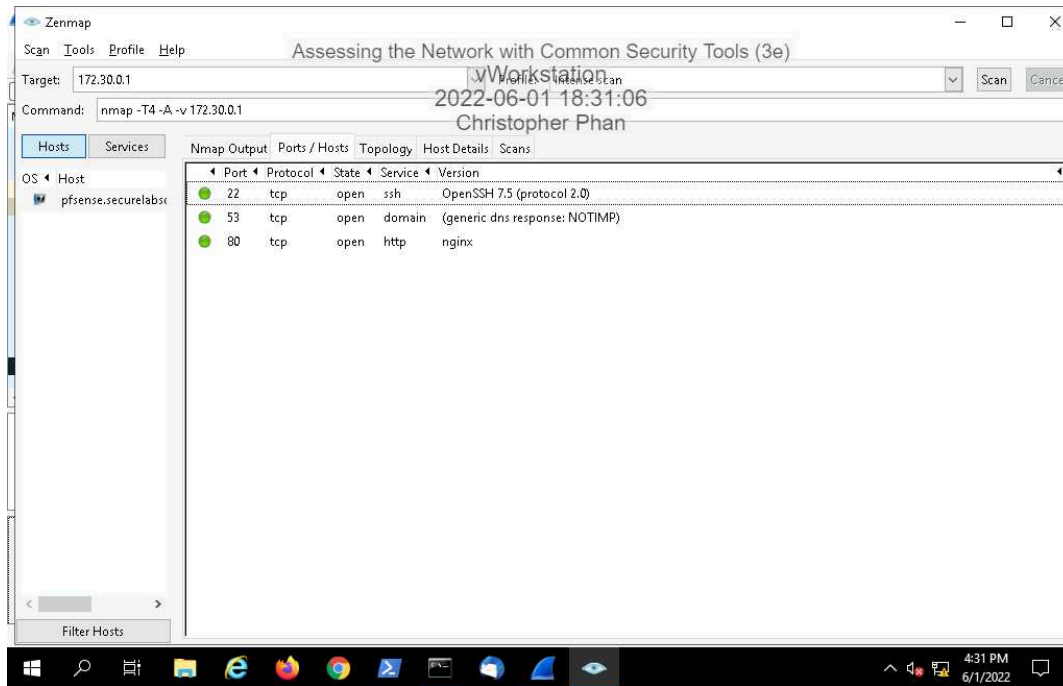
24. Compare the Intense scan results with the results from the Ping scan.

The Intense Scan is highly more detailed than both of the previous scans, in that it shows all the previous TCP ports in much more detail, as in what kind of OS is used, fingerprints, if there is any unrecognized service that is returning data to the scan. It goes in-depth about the Domain Name System using Nmap Scripting Engine, and about what steps are shown to give the results of the scan. Both the Regular and Ping do not show the details of these scans nor how they are conducted.

Assessing the Network with Common Security Tools (3e)

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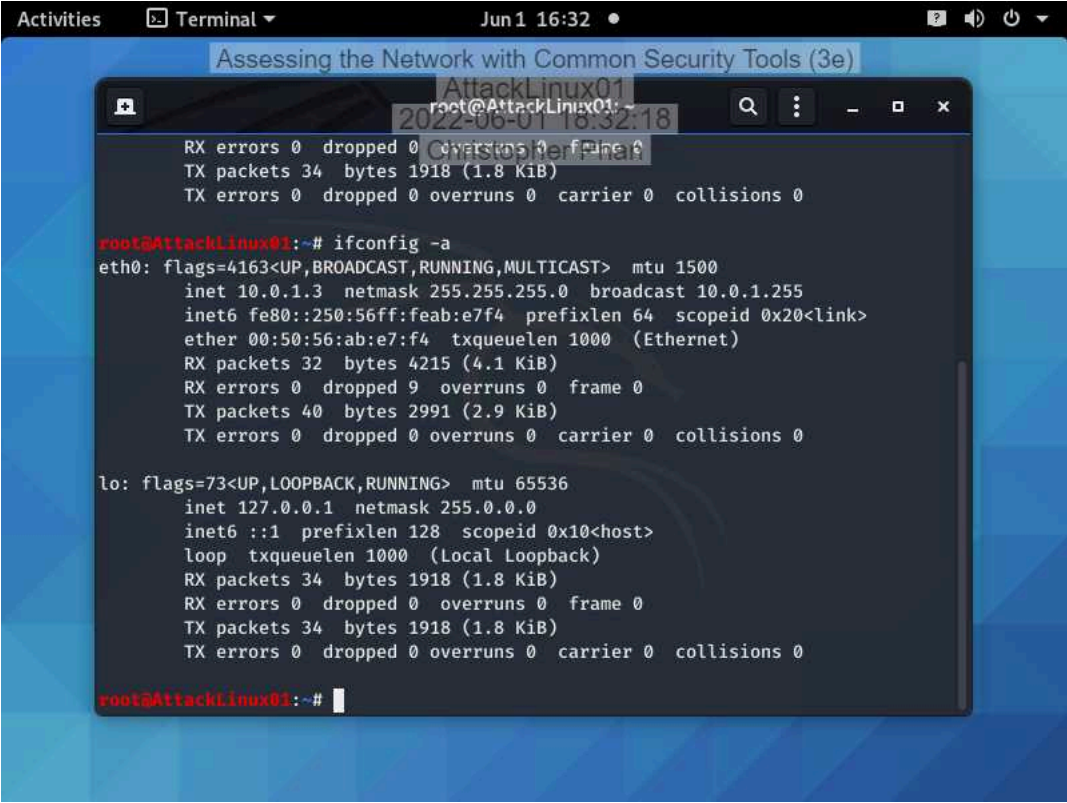
28. Make a screen capture showing the contents of the Ports/Hosts tab.



Section 2: Applied Learning

Part 1: Explore the Wide Area Network

6. Make a screen capture showing the `ifconfig` results on `AttackLinux01`.

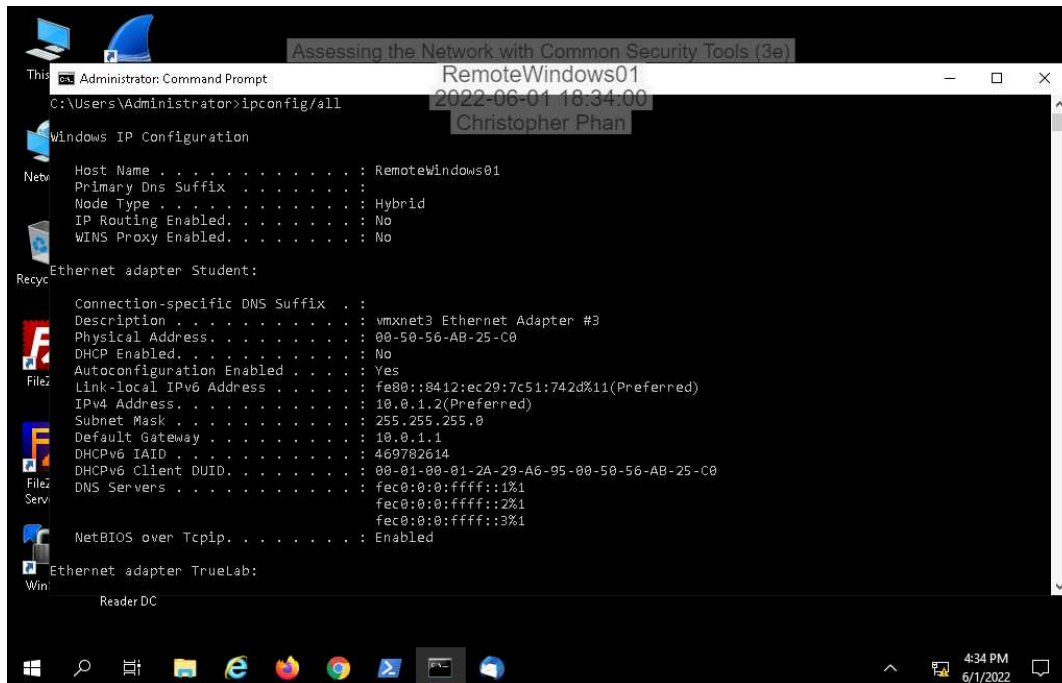


```
Activities Terminal Jun 1 16:32
Assessing the Network with Common Security Tools (3e)
AttackLinux01
root@AttackLinux01:~# ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.1.3 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fe80::250:56ff:feab:e7f4 prefixlen 64 scopeid 0x20<link>
    ether 00:50:56:ab:e7:f4 txqueuelen 1000 (Ethernet)
    RX packets 32 bytes 4215 (4.1 KiB)
    RX errors 0 dropped 9 overruns 0 frame 0
    TX packets 40 bytes 2991 (2.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

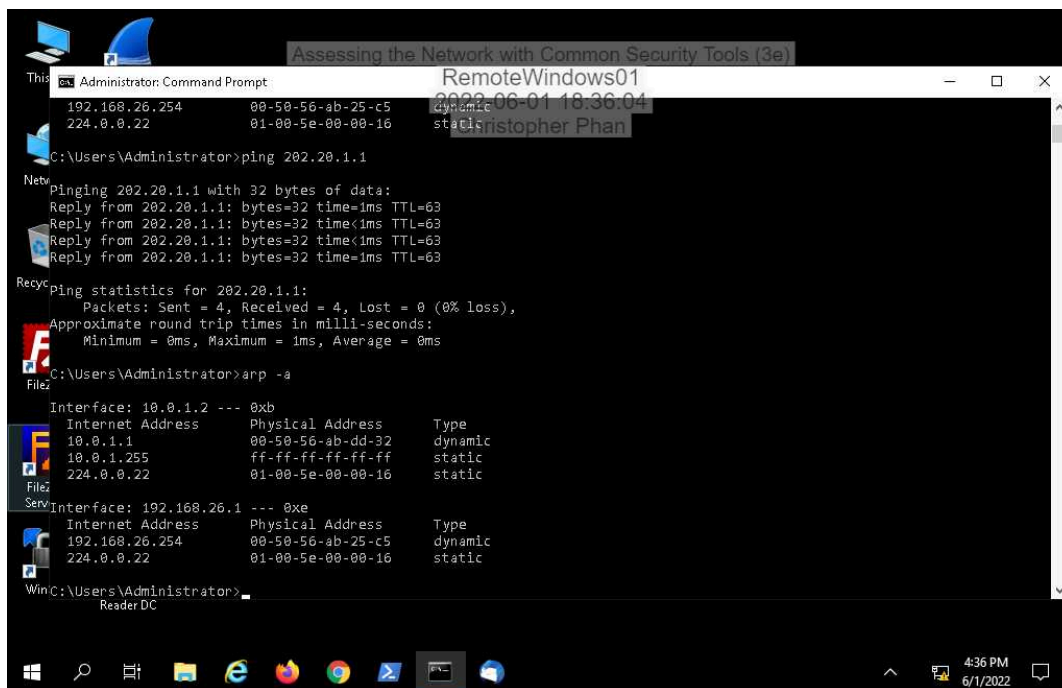
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 34 bytes 1918 (1.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 1918 (1.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@AttackLinux01:~#
```


12. Make a screen capture showing the ipconfig results on RemoteWindows01.



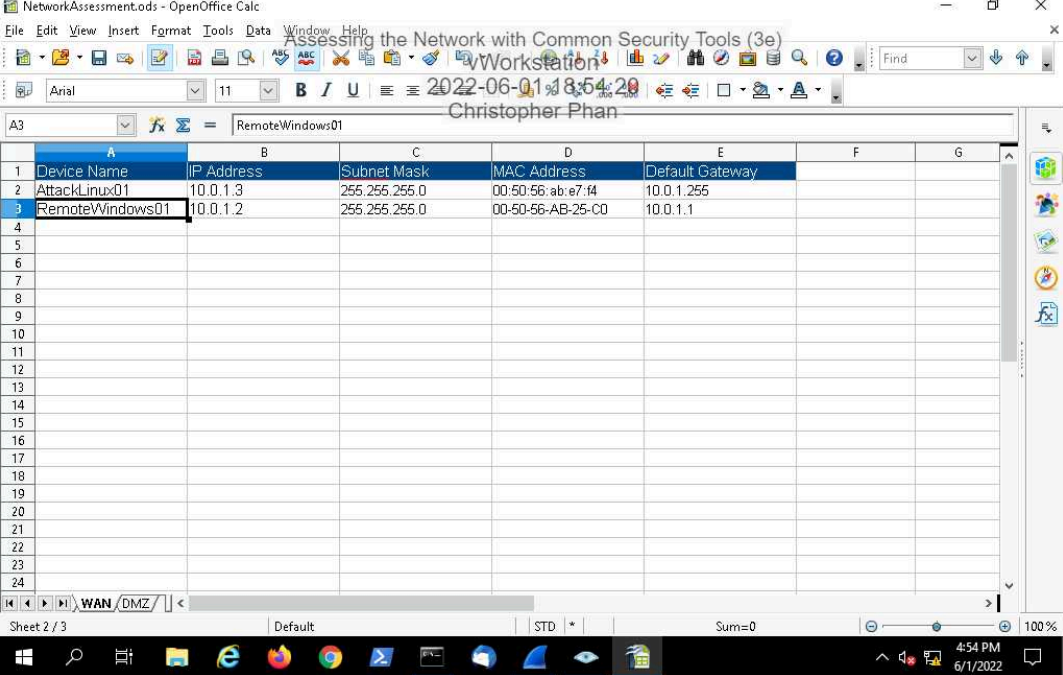
18. Make a screen capture showing the updated ARP cache on RemoteWindows01.



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22. Make a screen capture showing the **completed WAN tab** of the Network Assessment spreadsheet.



NetworkAssessment.ods - OpenOffice Calc

File Edit View Insert Format Tools Data Window Help

2022-06-01 18:54:20

Christopher Phan

	A	B	C	D	E	F	G
1	Device Name	IP Address	Subnet Mask	MAC Address	Default Gateway		
2	AttackLinux01	10.0.1.3	255.255.255.0	00:50:56:ab:e7:14	10.0.1.255		
3	RemoteWindows01	10.0.1.2	255.255.255.0	00:50:56:AB:25:C0	10.0.1.1		
4							
5							
6							
7							
8							
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18							
19							
20							
21							
22							
23							
24							

Sheet 2 / 3

Default

STD *

Sum=0

100 %

4:54 PM 6/1/2022

Part 2: Analyze Network Traffic

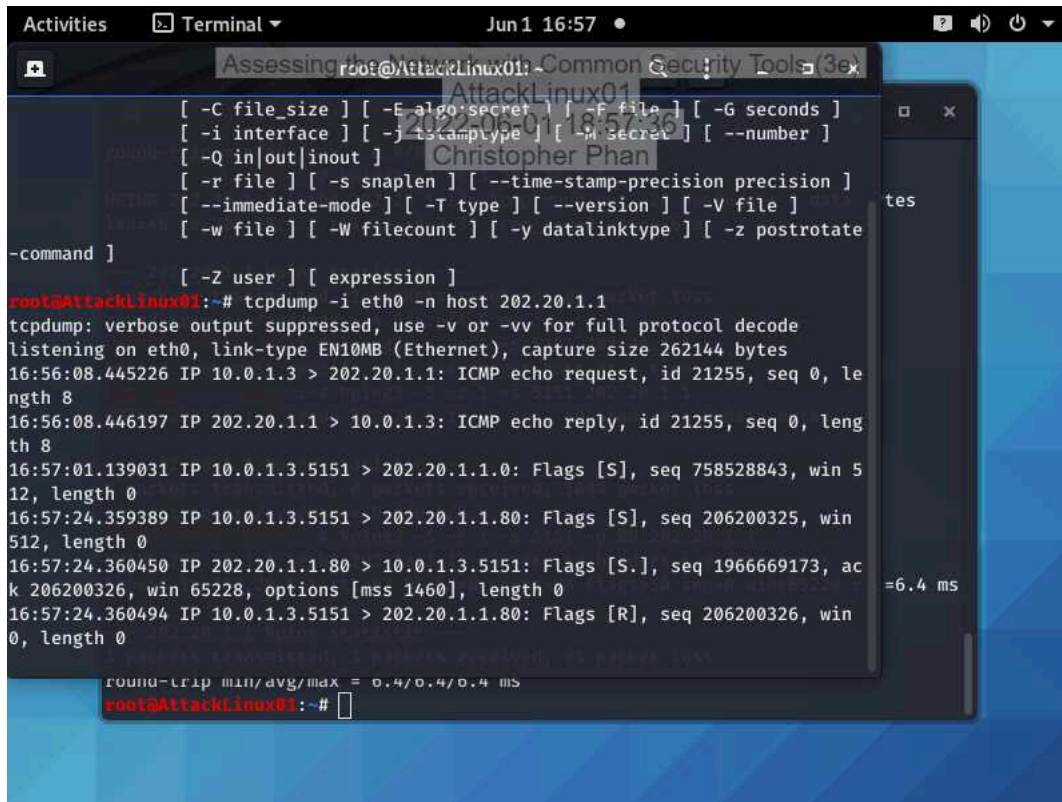
9. Make a screen capture showing **tcpdump** echo back the captured packets.

A screenshot of a Linux terminal window. The title bar shows 'Activities', 'Terminal', and the date/time 'Jun 1 16:56'. The terminal window has a title bar that says 'Assessing the Network with Common Security Tools (3e)'. The prompt is 'root@AttackLinux01:~#'. The user enters 'tcpdump -h', which shows the version (4.9.3), libpcap version (1.9.1), and OpenSSL version (1.1.1g). It then displays the usage of tcpdump with various options. The user then enters 'tcpdump -i eth0 -n host 202.20.1.1'. The output shows 'tcpdump: verbose output suppressed, use -v or -vv for full protocol decode', 'listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes', and two lines of captured traffic: '16:56:08.445226 IP 10.0.1.3 > 202.20.1.1: ICMP echo request, id 21255, seq 0, length 8' and '16:56:08.446197 IP 202.20.1.1 > 10.0.1.3: ICMP echo reply, id 21255, seq 0, length 8'. The prompt returns to 'root@AttackLinux01:~#'.

```
root@AttackLinux01:~# tcpdump -h
tcpdump version 4.9.3
libpcap version 1.9.1 (with TPACKET_V3)
OpenSSL 1.1.1g 21 Apr 2020
Usage: tcpdump [-aAbDefhHIJKlLnNopqStuUvxxX#] [-B size] [-c count]
        [-C file_size] [-E algo:secret] [-F file] [-G seconds]
        [-i interface] [-j tstamptype] [-M secret] [--number]
        [-Q in|out|inout]
        [-r file] [-s snaplen] [--time-stamp-precision precision]
        [--immediate-mode] [-T type] [--version] [-V file]
        [-w file] [-W filecount] [-y datalinktype] [-z postrotate]
        [-Z user] [expression]

root@AttackLinux01:~# tcpdump -i eth0 -n host 202.20.1.1
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
16:56:08.445226 IP 10.0.1.3 > 202.20.1.1: ICMP echo request, id 21255, seq 0, length 8
16:56:08.446197 IP 202.20.1.1 > 10.0.1.3: ICMP echo reply, id 21255, seq 0, length 8
root@AttackLinux01:~#
```

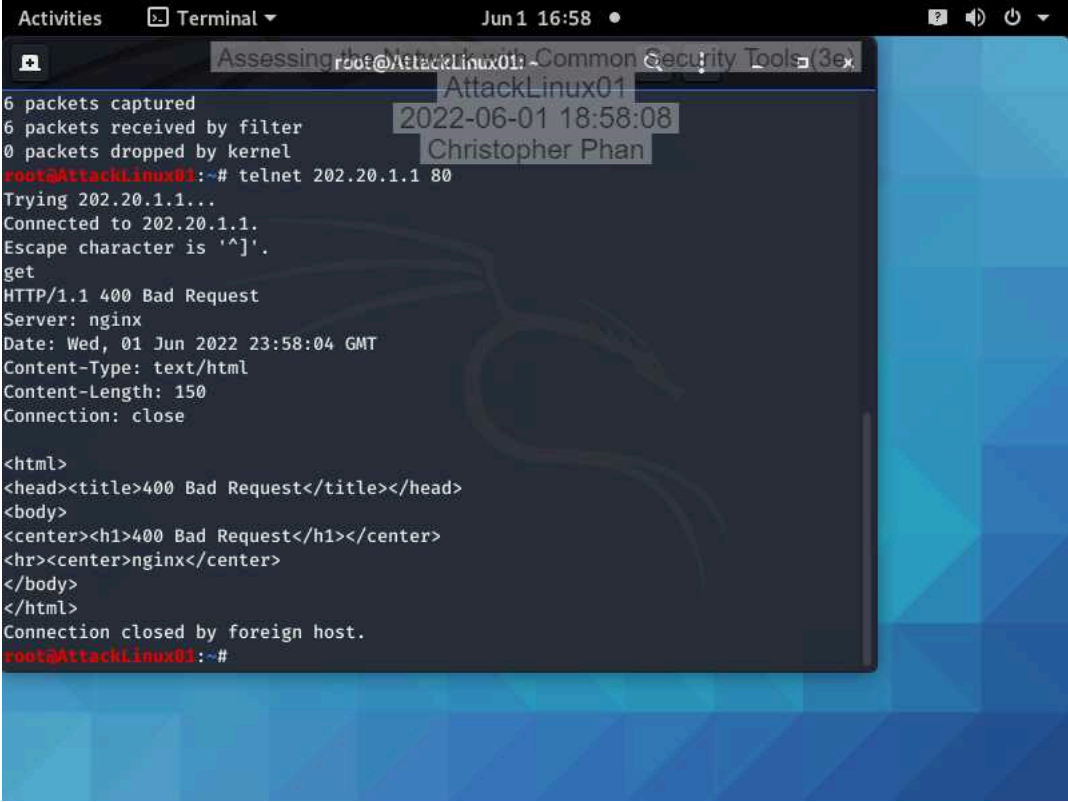
12. Make a screen capture showing the **attempted three-way handshake** in tcpdump.



```
root@AttackLinux01:~# tcpdump -i eth0 -n host 202.20.1.1
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
16:56:08.445226 IP 10.0.1.3 > 202.20.1.1: ICMP echo request, id 21255, seq 0, length 8
16:56:08.446197 IP 202.20.1.1 > 10.0.1.3: ICMP echo reply, id 21255, seq 0, length 8
16:57:01.139031 IP 10.0.1.3.5151 > 202.20.1.1.0: Flags [S], seq 758528843, win 512, length 0
16:57:24.359389 IP 10.0.1.3.5151 > 202.20.1.1.80: Flags [S], seq 206200325, win 512, length 0
16:57:24.360450 IP 202.20.1.1.80 > 10.0.1.3.5151: Flags [S.], seq 1966669173, ack 206200326, win 65228, options [mss 1460], length 0
16:57:24.360494 IP 10.0.1.3.5151 > 202.20.1.1.80: Flags [R], seq 206200326, win 0, length 0

round-trip min/avg/max = 0.4/0.4/0.4 ms
root@AttackLinux01:~#
```

17. Make a screen capture showing the results of the get command.



A terminal window titled "Assessing the Network with Common Security Tools (3e)" is shown. The window displays the output of a telnet session to 202.20.1.1 on port 80. The output shows a successful connection and the execution of the 'get' command, which returns an HTTP 400 Bad Request response from an nginx server. The response includes headers and an HTML body with a title and a message. The connection is then closed by the foreign host.

```
6 packets captured
6 packets received by filter
0 packets dropped by kernel
root@AttackLinux01:~# telnet 202.20.1.1 80
Trying 202.20.1.1...
Connected to 202.20.1.1.
Escape character is '^]'.
get
HTTP/1.1 400 Bad Request
Server: nginx
Date: Wed, 01 Jun 2022 23:58:04 GMT
Content-Type: text/html
Content-Length: 150
Connection: close

<html>
<head><title>400 Bad Request</title></head>
<body>
<center><h1>400 Bad Request</h1></center>
<hr><center>nginx</center>
</body>
</html>
Connection closed by foreign host.
root@AttackLinux01:~#
```

Section 3: Challenge and Analysis

Part 1: Explore the DMZ

Make a screen capture showing the **completed DMZ tab** of the **NetworkAssessment** spreadsheet.

Device Name	IP Address	Subnet Mask	MAC Address	Default Gateway
TargetLinux01	172.40.0.20	255.255.255.0	00:50:56:ab:60:d3	172.40.0.255

Part 2: Perform Reconnaissance on the Firewall

Briefly summarize and analyze your findings in a technical memo to your boss.

There were 2 ICMP packets sent to the firewall. There are two ports, port 22 SSH and 80 HTTP.